Dhruv Chavan

Phone Number: (614) 705-9999 | Email: dhruvchavan59@gmail.com | LinkedIn

EDUCATION

The Ohio State University **Expected Graduation: Spring 2027**
B.S. in Data Analytics - *Honors and Scholars Student* GPA: 3.66/4.0

PUBLICATIONS

- Giuliani G., **Chavan D.**, Gaddam R., Stewart W.C., Li Z., Jayaprakash C., Das J. "Immune and non-immune cell fencing of tumor cells is a widespread and functionally relevant spatial pattern in solid cancers." *Computational and Structural Biotechnology Journal*, 2025.
- Diawara J., **Chavan D.**, Frey G., DiGiovine C., Mhatre A. "Analysis of functional mobility assessment outcomes measures at a wheelchair clinic." *RESNA Annual Conference*, 2025.

RELEVANT COURSEWORK

Aerospace & Controls: Guidance Navigation & Controls, Systems Linear Optimization and Modeling
Statistics & ML: Probability, Statistical Inference, Statistical Modeling Discovery I & II, Statistical Decision Making, Statistical Learning, Artificial Intelligence Survey, and Data Mining
Computer Science: Programming Python, Software I & II, Discrete Structures, Low Level Programming, Database Systems, Data Structures & Algorithms, Data Visualization

RESEARCH EXPERIENCE

Controls and Navigation Research Lead **January 2025 - Present**
*Department of Mechanical and Aerospace Engineering at OSU - **Dr. Mrinal Kumar*** *Columbus, OH*

- Architecting autonomous wildfire drones from scratch, calibrating Pixhawk with fused RGB/IR sensor streams
- Deploying CNN segmentation on Jetson Nano to classify fire-risk terrain at **30ms** and **92%** accuracy
- Validating flight and mission logic in pymavlink SITL simulation, cutting field-test iteration cycles
- Leading **5** students building ground robots that relay data to drones and ignite controlled test fires
- Building multi-node ROS2 stack for LiDAR, AI camera, and smoke sensors; wrote custom camera drivers
- Achieving **2cm** aerial and **5cm** ground accuracy via Marvelmind indoor GPS with Nav2 path planning

Aerospace Student Researcher
*Department of Mechanical and Aerospace Engineering at OSU - **Dr. Ali Jhemi***

- Wrote a Kalman filter from scratch in MATLAB, C/Arduino, and Python, cutting orientation error by **85%**
- Designed and 3D-printed a 3-DOF IMU stand in Fusion 360 with encoder ground truth; accepted at AIAA 2026
- Developing an active sidestick for a Honda Aircraft simulator: **≤10ms** latency, **500Hz** force feedback

Undergraduate Student Researcher **July 2024 - Present**
*OSU Wexner Medical Center - **Dr. Carmen DiGiovine*** *Columbus, OH*

- Published research as second author at the international RESNA conference and attended the 2025 Rehabweek
- Cleaning outpatient clinical data from assistive technology center to ensure analysis-ready accuracy
- Analyzing **4,000+** patient data points; findings presented to clinicians driving improvement in outcomes
- Applying hypothesis testing to identify demographic correlations and create visual representations
- Writing literature reviews to support research papers and contextualize data findings

Undergraduate Computational Biology Researcher **February 2024 - June 2025**
*Nationwide Children's Hospital - **Dr. Jayajit Das*** *Columbus, OH*

- Second author on peer-reviewed CSBJ paper analyzing tumor structures across cell types
- Developed advanced visualizations to analyze brain cancer cell interactions with immune cells
- Analyzed **1.2M+** spatial data points from **30+** tumors, revealing cell clustering linked to ICI response
- Built permutation-testing pipeline running **10,000+** randomized trials per slide to test significance

INDUSTRY EXPERIENCE

Summer Analyst Intern **June 2026 - July 2026**
JPMorganChase *Columbus, OH*

- Automated vendor performance reporting in Snowflake, streamlining analytics delivery to senior executives
- Conducted risk assessments on new initiatives to detect fraud patterns and support compliance reporting
- Built authentication risk dashboards to surface fraud insights and monitor performance within tolerance

Data and AI Engineering Intern **September 2025 - December 2025**
BMW Financial Services *Columbus, OH*

- Built MCP server tools connecting internal LLMs to PostgreSQL, improving data retrieval efficiency by **45%**
- Developed **3** Gen AI apps spanning fraud detection, business intelligence, and call center automation
- Launched an AI governance platform enabling **95+** employees to submit model requests for compliance protocols

PROJECTS

President's Buckeye Accelerator Program - *Keenan Center* **January 2025 - May 2025**

- Developed a YOLO-based computer vision prototype assisting 5 test users with facial recognition deficits
- Conducted user research through surveys and video calls; **95%** of respondents validated the idea
- Selected through competitive workshops focused on market strategy, design, and entrepreneurship

Autonomous Lego Technic Bugatti **May 2026**

- Added custom hardware to pre existing chassis with an onboard raspberry pi for computing
- Reduced path deviation by **75%** fusing IMU with camera via Extended Kalman Filter over camera-only
- Achieved **80%** lane-following accuracy via adaptive thresholding at **30fps**, publishing steering at **20Hz**
- Architected **7-node** ROS2 Jazzy stack separating perception, planning, and safety layers mirroring AV stacks
- Engineered safety node stopping the vehicle within **20cm**; achieved **76%** parallel parking success over **25** trials

Animal Image Classification ML Model **June 2024**

- Designed an object-oriented CNN using TensorFlow and Keras to classify animal images with **98.2%** accuracy
- Trained on a balanced dataset of **5** animal classes, applying data augmentation to improve robustness.
- Optimized performance with multiple convolutional and pooling layers to enhance feature extraction and accuracy

SKILLS

- **Languages:** Python, C++, C, MATLAB, R, SQL
- **ML/Perception:** PyTorch, YOLO, OpenCV, scikit-learn, Pandas, NumPy
- **Robotics:** ROS2, Nav2, Pixhawk/PX4, pymavlink, Kalman filtering, sensor fusion
- **Hardware/Platforms:** Jetson Nano, Raspberry Pi, Arduino, BNO055 IMU, LiDAR, Marvelmind
- **Tools/Frameworks:** Git/GitHub, CMake, Docker, Fusion 360

LEADERSHIP

Calculus Tutor **January 2025 - August 2025**
Student-Athlete Support Services Office at OSU *Columbus, OH*

- Tutoring 3 student-athletes weekly (~10 hrs/week) in Calculus I & II around tight travel schedules
- Developing and implementing personalized lesson plans tailored to individual learning needs
- Received consistently positive feedback and biweekly performance evaluations from the athletics office

Mount Scholars Leadership Society **August 2023 - Present**

- Led Team Security initiatives and developed leadership through weekly training, mentorship, and retreats
- Invited to a Washington D.C. trip and learned from leaders from the Department of Justice and Treasury
- Mentoring underclassmen as a designated "Mega," providing guidance on academics and early college transition
- Completed 75+ hours mentoring athletes with Down syndrome through weekly tennis sessions and events
- Attending monthly service projects, including local river cleanups and meal preparation for a food pantry